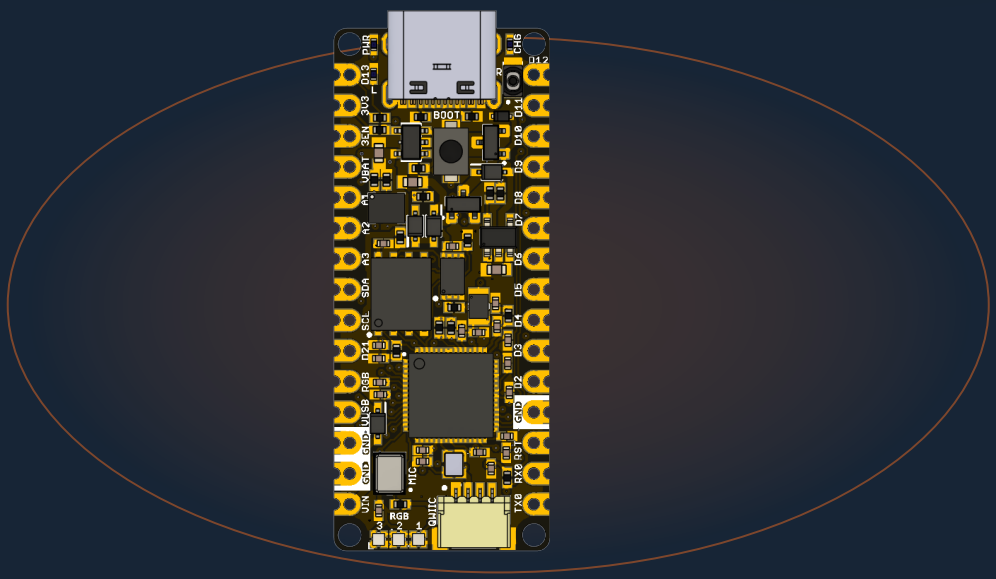
PRODUCT REFERENCE MANUAL

MEMBER OF THE UNIT DEVLAB ECOSYSTEM

# UNIT PULSAR RP2350 SENSE

**MPN: UE0103**

Version 0.1.0




RP2350A development board with external flash and PSRAM, motion sensing, PDM audio, microSD, QWIIC, USB-C, and HSTX expansion.

RP2350A | 16 MiB FLASH | 8 MiB PSRAM | SDIO · PDM · I2C · HSTX

DOCUMENT MAP

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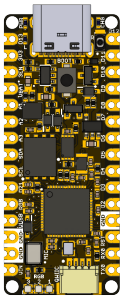
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# Description

The UNIT PULSAR RP2350 is a multi-interface development board in the UNIT DevLab ecosystem, built around the Raspberry Pi RP2350A microcontroller. The V1.3 design combines external QSPI flash and PSRAM with motion sensing, PDM audio, microSD, USB-C, battery support, addressable RGB indicators, QWIIC I2C, and a 22-pin HSTX expansion connector.



## Applications

- RP2350 firmware and peripheral prototyping
- Motion and PDM audio acquisition
- microSD data logging
- HSTX graphics experiments
- External-memory and multicore applications
- Education and subsystem validation

## Hardware Features

- RP2350A microcontroller with external W25Q128 128 Mbit QSPI flash
- APS6404L-3SQR-ZR 8 MiB PSRAM
- BMI270 six-axis inertial measurement unit
- ICS-41350 digital PDM microphone
- Four-bit SDIO microSD connection
- Three WS2812-compatible RGB LEDs plus power, charge, and user indicators
- USB-C, QWIIC, battery, edge-pad, SWD, and HSTX connections

# 1 The Board

The UNIT PULSAR RP2350 is a development board for evaluating the Raspberry Pi RP2350A and building applications that combine high-speed digital output, removable storage, motion sensing, digital audio, external memory, and general-purpose I/O. Hardware revision V1.3.0 places the controller, QSPI flash, PSRAM, power components, indicators, and user controls on the top side. The microSD socket, battery connection, onboard microphone, debug pads, and HSTX connector are accessible from the bottom side.

## 1.1 Accessories

The following optional accessories are recommended for use with the UNIT PULSAR RP2350A. Select accessories according to the interfaces and features required by the application.

| Accessory | Purpose | Selection notes |
|---|---|---|
| QWIIC cable | External I2C expansion | 4-pin, 1 mm pitch cable for connecting compatible 3.3 V QWIIC devices |
| USB-C data cable | Power, programming, and USB serial | Must support data; a charge-only cable cannot upload firmware |
| MicroSD card | Removable storage and data logging | Class 10 microSD card; FAT32 formatting is recommended for the provided examples |
| UNIT DevLab DVI to FPC Adapter | HSTX DVI-compatible video output | Recommended adapter for using the board's HSTX video interface through the 22-pin FPC connector |
| 22-pin FPC cable | Connection between the HSTX connector and DVI adapter | 22-position, 0.5 mm pitch; planned for future availability from UNIT Electronics |
| LiPo 3.7 V 650 mAh Battery | Battery-powered operation | Recommended single-cell LiPo battery; verify polarity and connector orientation before connection |

Verify connector type, orientation, polarity, voltage domain, and pinout before connecting accessories.

## 1.2 Board Identification

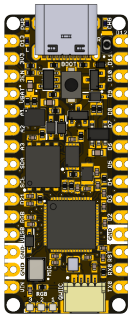
| Item | Value |
|---|---|
| Product | UNIT PULSAR RP2350 |
| Product family | UNIT DevLab ecosystem |
| Product type | Multi-interface RP2350A development board |
| Main component | Raspberry Pi RP2350A, QFN-60 |
| Hardware revision | V1.3.0 |
| Product Reference | Version 0.1.0 |
| Primary programming interface | USB-C / RP2350 USB boot workflow |
| Debug interface | SWD pads on bottom side |

Hardware revision and documentation revision are identified independently.

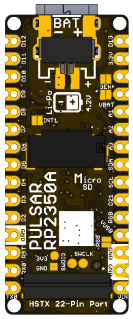
## 1.3 Main Assemblies

| RefDes | Component | Function |
|---|---|---|
| IC3 | RP2350A | Main processor and peripheral controller |
| IC1 | W25Q128JVPIQ | 128 Mbit (16 MiB) external QSPI flash |
| IC4 | APS6404L-3SQR-ZR | 8 MiB external PSRAM |
| IC5 | BMI270 | Six-axis accelerometer and gyroscope |
| MK1 | ICS-41350 | Digital PDM microphone |
| U1 | AP2112K-3.3TRG1 | Fixed 3.3 V LDO |
| IC2 | MCP73831T-2ACI/OT | Single-cell Li-Ion/Li-Polymer charge controller |
| XTAL1 | XOS20012000LT00351005 | 12 MHz reference oscillator |
| MICRO_SD-HOLDER | 47309-2651 | Removable microSD storage |
| LED1–LED3 | WS2812 1010 | Cascaded addressable RGB indicators |
| J1 | HCZZ0032-4 | Four-position QWIIC-style I2C connector |
| J5 | FH34SRJ-22S-0.5SH(50) | 22-position HSTX FFC/FPC connector |
| JP1 | PH2.0 2P | Two-position battery connection |

## 1.4 Board Views



The top view identifies the USB-C connector, BOOT and reset controls, edge-pad labels, RP2350A, flash, PSRAM, QWIIC connector, oscillator, and three RGB LEDs.




The bottom view identifies the battery polarity marks, microSD socket, microphone acoustic port, SWD pads, HSTX connector, and V1.3.0 revision marking.

## 1.5 Handling

Handle the board using normal ESD precautions. Avoid touching the microphone port, connector contacts, or exposed test pads. Remove power before inserting or removing FFC/FPC and battery connectors. Keep conductive objects away from the bottom-side battery and microSD areas when the board is energized.

# 2 Ratings

This chapter defines the recommended operating conditions and electrical characteristics of the UNIT PULSAR RP2350A hardware V1.3.0. Ratings apply to the complete board unless otherwise stated.

## 2.1 Recommended Operating Conditions

The UNIT PULSAR RP2350A can be powered from USB-C, VIN, or a single-cell LiPo battery. The onboard AP2112K regulator supplies the 3.3 V logic domain used by the RP2350A and onboard peripherals.

| Parameter | Recommended operating condition | Notes |
|---|---|---|
| USB-C input | 5 V nominal | Primary power and programming interface |
| VIN input | Up to 6 V | External board supply input |
| Battery input | 3.7 V nominal, 4.2 V maximum | Single-cell LiPo battery |
| Regulated logic rail | 3.3 V nominal | Generated by the onboard AP2112K-3.3 regulator |
| 3.3 V regulator output capability | Up to 600 mA | Shared by the RP2350A, onboard peripherals, LEDs, and external 3.3 V loads |
| Battery charge current | 200 mA nominal | Configured by the MCP73831 PROG resistor |
| RP2350 system clock | Up to 150 MHz | Firmware configurable |
| Board reference oscillator | 12 MHz | Onboard reference oscillator |
| Internal SRAM | 520 kB | RP2350A internal memory |
| External flash | 128 Mbit / 16 MiB | W25Q128JVPIQ |
| External PSRAM | 8 MiB | APS6404L-3SQR-ZR |
| Logic level | 3.3 V | Applies to onboard digital interfaces and GPIO |

## 2.2 Power Domains and Power Distribution

The UNIT PULSAR RP2350A uses separate power domains for external supply, battery operation, system power, and regulated 3.3 V logic.

The main power domains are:

- **USB-C / VBUS:** provides the nominal 5 V USB supply used for board power, programming, and battery charging.
- **VIN:** external supply input supporting up to 6 V.
- **VBAT:** battery domain intended for a single-cell LiPo battery with a nominal voltage of 3.7 V and a maximum fully charged voltage of 4.2 V. The battery input includes reverse-polarity protection.
- **VSYS:** internal system supply used as the input to the 3.3 V regulator. The board power-selection circuitry routes the available supply source to this domain.

- **3.3 V:** regulated logic supply generated by the AP2112K-3.3. This rail powers the RP2350A, external flash, PSRAM, BMI270, PDM microphone, microSD interface, QWIIC interface, RGB LEDs, and other onboard 3.3 V circuitry.

The AP2112K-3.3 regulator supports up to 600 mA output current. This current is shared between the RP2350A, onboard peripherals, status and RGB LEDs, and any external device powered from the 3.3 V rail. The current available to external loads therefore depends on the active board configuration and application workload.

The MCP73831T-2ACI/OT manages single-cell LiPo battery charging. Hardware V1.3.0 is configured for a nominal charge current of 200 mA. The charging current is determined by the PROG resistor and may be changed by modifying its value according to the MCP73831 configuration requirements.

All onboard digital interfaces operate in the 3.3 V logic domain. QWIIC, microSD, HSTX, and exposed GPIO signals must therefore be used with 3.3 V-compatible devices unless external level translation is provided.

Connector mechanical current ratings must not be interpreted as board power ratings. Maximum usable current is also constrained by the associated PCB traces, power path, regulator, protection components, and operating conditions.

## 2.3 Digital Interfaces

All onboard digital peripherals are connected to the board's 3.3 V logic domain. External logic compatibility must nevertheless be checked against the RP2350A input/output specifications and the power state of both devices.

| Interface | Board connection | Electrical note |
|---|---|---|
| Internal I2C | GPIO8 / GPIO9 | BMI270 bus with onboard pull-ups |
| QWIIC I2C | GPIO24 / GPIO25 | Connector also supplies 3.3 V and GND |
| microSD | GPIO2–GPIO7 | Full SDIO group; wiki uses SPI-compatible subset |
| PDM | GPIO10 / GPIO11 | Clock output and microphone data input |
| HSTX | GPIO12–GPIO19 | High-speed driven pairs; reserve while video is active |
| WS2812 data | GPIO1 | Drives three cascaded onboard LEDs |
| SWD | SWDIO / SWCLK | Use a 3.3 V target reference |

## 2.4 Confirmed Component Values

| Symbol / parameter | Value | Scope |
|---|---:|---|
| RP2350 internal SRAM | 520 kB | RP2350A component |
| RP2350 system clock | 150 MHz maximum nominal | RP2350A component |
| QSPI flash capacity | 128 Mbit | IC1 |
| PSRAM capacity | 8 MiB | IC4 |
| PDM microphone supply | 1.65 to 3.63 V | ICS-41350 component only |
| LDO output | 3.3 V nominal | U1 component selection |
| Oscillator frequency | 12 MHz | XTAL1 |

## 2.5 Application-Dependent and Uncharacterized Parameters

Some electrical and performance characteristics depend on firmware configuration, operating mode, connected peripherals, and application load. Unless otherwise specified, these parameters should not be interpreted as guaranteed board-level limits.

- **Available 3.3 V current for external loads:** the AP2112K-3.3 provides up to 600 mA total output current. This current is shared by the RP2350A, onboard memories, sensors, LEDs, and external 3.3 V loads. Available external current therefore depends on the active board configuration and application workload.
- **Board current consumption:** total current consumption varies with RP2350A activity, clock configuration, enabled peripherals, RGB LED brightness, external memory activity, microSD operation, audio capture, HSTX video output, and connected external devices.
- **High-speed interface performance:** maximum validated operating rates for external microSD, PDM, I2C, and HSTX interfaces depend on firmware configuration, connected devices, cabling, and signal integrity.
- **GPIO electrical characteristics:** GPIO drive strength, input thresholds, and other pin-level electrical characteristics follow the RP2350A specifications unless a board-level circuit modifies the corresponding signal.
- **Environmental characteristics:** board-level operating temperature, humidity, ESD immunity, and thermal performance are not specified unless explicitly stated in a released qualification or validation specification.

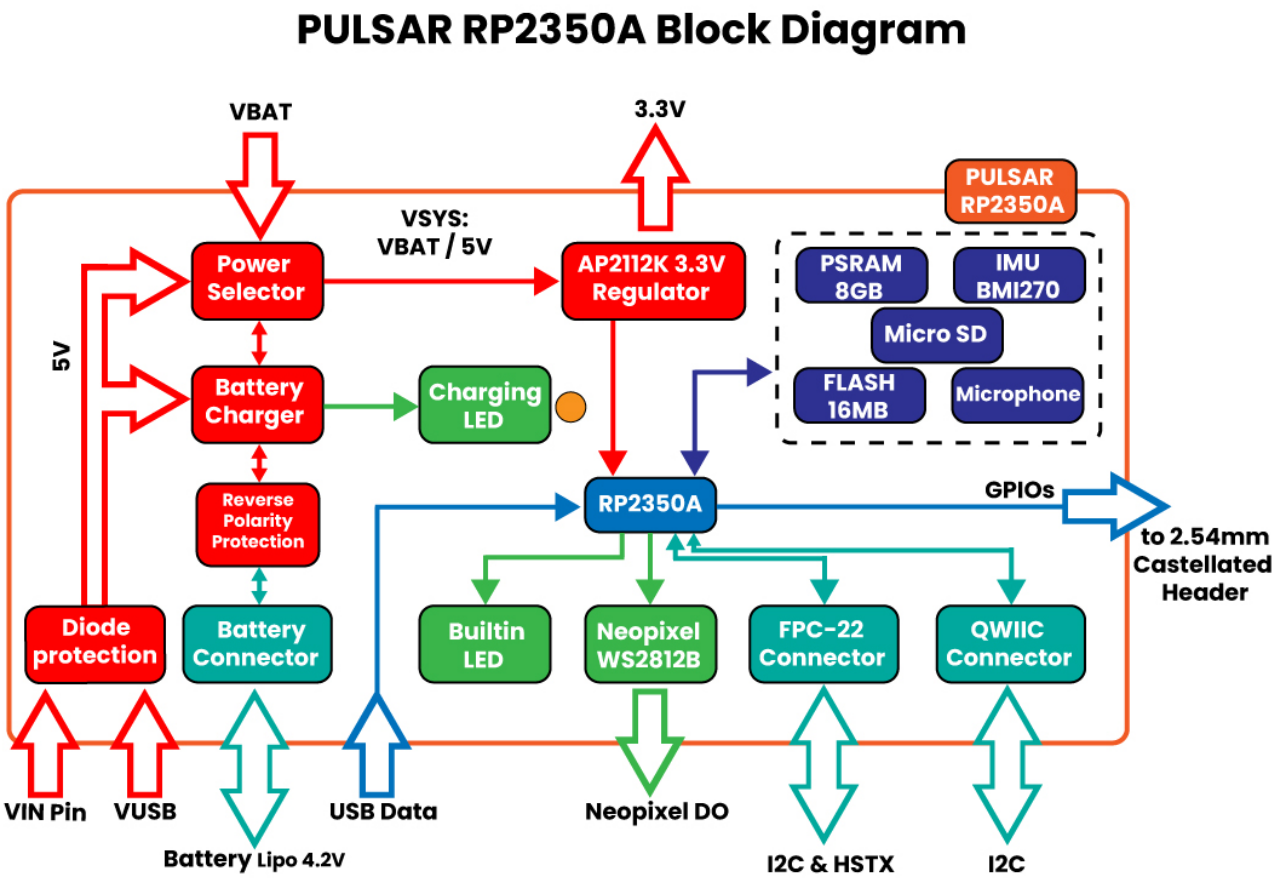
## 2.6 Electrical Precautions

1. Use only the specified power-input ranges: 5 V nominal through USB-C, up to 6 V through VIN, or a 3.7 V nominal / 4.2 V maximum single-cell LiPo battery.
2. All GPIO and onboard digital interfaces operate in the 3.3 V logic domain. **Do not apply 5 V directly to GPIO, QWIIC, microSD, HSTX, SWD, or other 3.3 V signal lines.**
3. Observe the battery connector polarity before connecting a single-cell LiPo battery. Do not connect multi-cell battery packs or batteries exceeding 4.2 V.
4. Do not externally drive the 3.3 V rail while the onboard regulator is active unless the external power source and operating configuration have been specifically designed for this purpose.
5. Connect external circuits to a common ground before applying external clock, data, or control signals.
6. Do not connect externally driven circuits to GPIO12–GPIO19 while the HSTX interface is active, as these GPIOs are used by the HSTX video output.
7. Remove power before inserting, removing, or changing FFC/FPC or battery connections. Avoid inserting or removing the microSD card while a write operation is in progress.
8. Ensure that the total load on the 3.3 V rail, including onboard circuitry and externally connected devices, remains within the 600 mA capability of the AP2112K-3.3 regulator.

# 3 Functional Overview

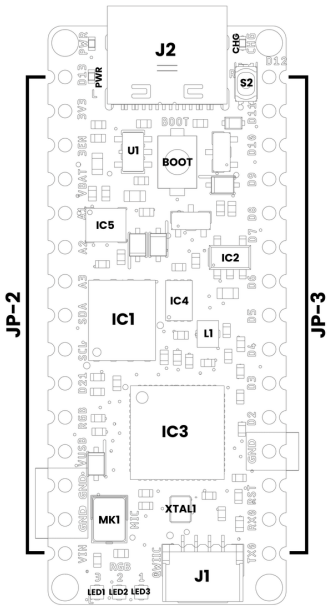
The UNIT PULSAR RP2350 is organized around the RP2350A processor, with dedicated external memories and fixed onboard peripherals. Firmware can use each block independently or combine acquisition, storage, user feedback, and video in one application.

## 3.1 Block Diagram

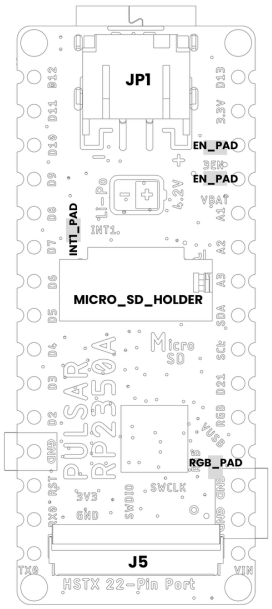



The RP2350A controls every data path. W25Q128 flash stores executable code and nonvolatile application data. PSRAM extends working memory. The microSD socket provides removable storage. BMI270 and ICS-41350 devices provide motion and audio input. QWIIC and edge pads expose expansion signals, while HSTX and the LEDs provide high-speed and visual outputs.

## 3.2 Board Topology




Top Side




Bottom Side

# Views of Board Topology

| RefDes | Component | Confirmed role |
|---|---|---|
| IC3 | RP2350A | Main microcontroller |
| IC1 | W25Q128JVPIQ | 128 Mbit (16 MiB) QSPI flash memory |
| IC4 | APS6404L-3SQR-ZR | 8 MiB external PSRAM |
| IC5 | BMI270 | Six-axis IMU connected to the internal I2C bus |
| MK1 | ICS-41350 | Digital PDM MEMS microphone |
| U1 | AP2112K-3.3TRG1 | Fixed 3.3 V LDO regulator |
| IC2 | MCP73831T-2ACI/OT | Single-cell Li-Ion/LiPo battery charge controller |
| XTAL1 | 12 MHz oscillator | RP2350A reference clock source |
| L1 | 3.3 µH inductor | RP2350A internal switching-regulator inductor |
| MICRO_SD_HOLDER | 47309-2651 | microSD card socket connected through four-bit SDIO signals |
| LED1–LED3 | XL-1010RGBC-WS2812B | Cascaded addressable RGB LEDs |

| RefDes | Component | Confirmed role |
|---|---|---|
| J1 | HCZZ0032-4 | Four-position, 1 mm pitch QWIIC connector |
| J2 | USB-C connector | USB power, programming, and USB data interface |
| JP1 | Battery connector | Two-position single-cell LiPo battery connection |
| J5 | FH34SRJ-22S-0.5SH(50) | 22-position, 0.5 mm pitch FFC/FPC connector for HSTX video and auxiliary signals |
| S2 | BOOT switch | Forces RP2350A USB boot mode during startup/reset |
| JP2 | Castellated header row | Left-side GPIO, power, analog, and digital expansion |
| JP3 | Castellated header row | Right-side GPIO, power, analog, and digital expansion |
| EN_PAD | Solder jumper | Connects the AP2112K enable signal to the castellated header, allowing external control of the 3.3 V regulator |
| VBAT_PAD | Solder jumper | Connects the VBAT rail to the castellated header, providing access to the battery supply |
| INT_PAD | Solder jumper | Connects the BMI270 interrupt signal to the RP2350A GPIO, enabling hardware interrupt operation |
| RGB_PAD | Solder jumper | Connects the data output of the third onboard NeoPixel to the castellated header, allowing the RGB chain to be extended with external NeoPixels |
| SWDIO | Debug test pad | RP2350A Serial Wire Debug data signal |
| SWCLK | Debug test pad | RP2350A Serial Wire Debug clock signal |
| GND | Debug test pad | Ground reference for the SWD interface |

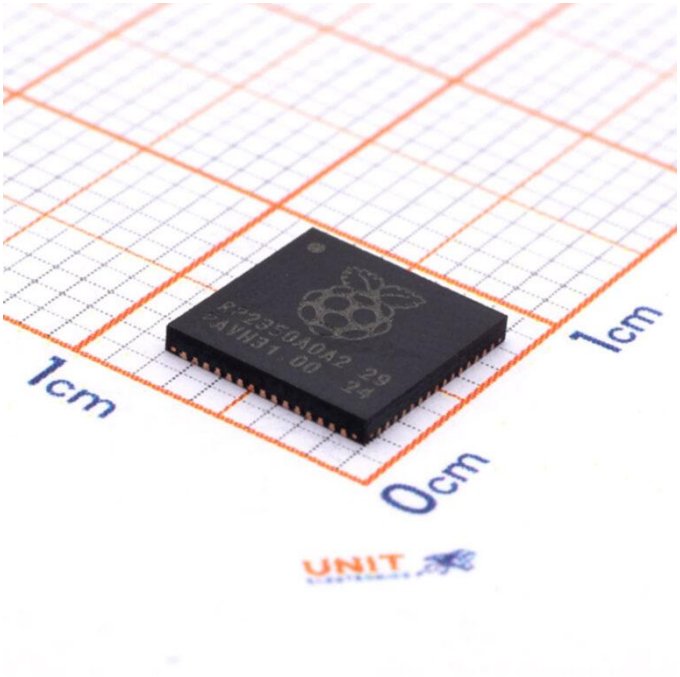
The top side groups programming, processing, memory, and user-visible controls. USB-C, BOOT, reset, QSPI flash, PSRAM, RP2350A, QWIIC, and RGB indicators are visible from this side. The bottom side groups removable storage, battery, audio, SWD, and HSTX connections.

| Board region | Main elements | Design intent |
|---|---|---|
| USB end | USB-C, BOOT, reset, power/charge indicators | Programming and initial power-up |
| Upper center | power-path devices and PSRAM | Source routing and working-memory expansion |
| Center | RP2350A and 12 MHz oscillator | Processing and clock generation |
| Lower center | QSPI flash and PDM microphone area | boot storage and audio acquisition |
| QWIIC end | three RGB LEDs and QWIIC connector | status display and external I2C |
| Bottom side | battery, microSD, BMI270, SWD, HSTX | storage, sensing, debug, and video expansion |

The two edge-pad rows expose power, analog, serial, and general-purpose signals. Some HSTX-routed signals are shared with edge positions, so firmware ownership must be decided before enabling high-speed video.

## 3.3 Processor

RP2350A is the main controller. According to the Raspberry Pi component datasheet, the device provides two selectable processor architectures: dual Arm Cortex-M33 or dual Hazard3 RISC-V cores, with a nominal system frequency up to 150 MHz. Firmware selects one architecture for a given build; it does not execute Arm and RISC-V code simultaneously.




*Raspberry Pi RP2350A QFN-60 microcontroller*

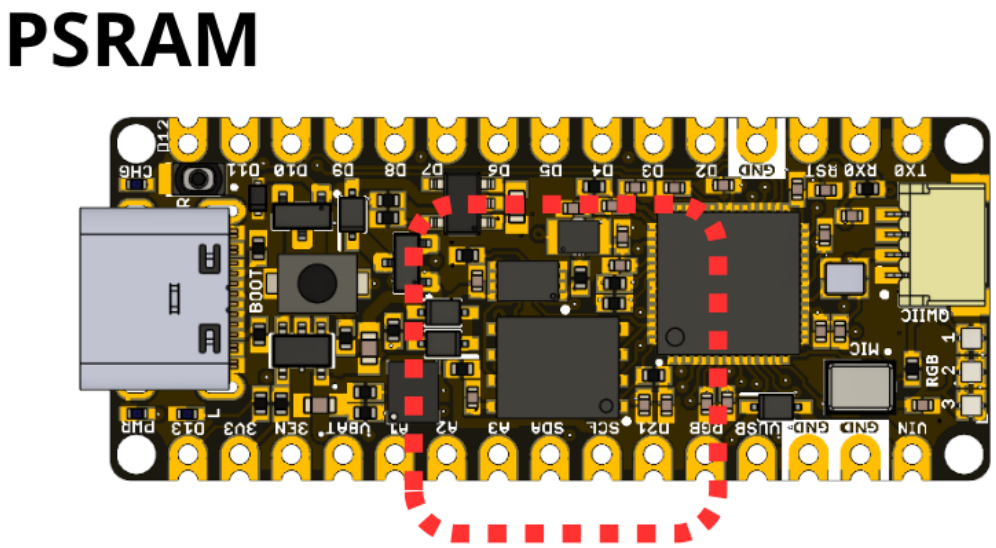
The processor integrates 520 kB of on-chip SRAM arranged in ten banks, USB 1.1 host/device control with an embedded PHY, ADC inputs shared with GPIO, fixed serial peripherals, PWM, DMA, three PIO blocks with twelve state machines, and one HSTX peripheral. The PULSAR design routes a selected subset of these functions to onboard devices and external connectors.

The on-chip SRAM is the lowest-latency working memory. It contains stacks, heaps, interrupt data, time-critical buffers, and code copied from flash when a framework requires it. DMA and multicore applications must coordinate access to shared buffers and peripherals.

## 3.4 Memory Architecture: Flash and PSRAM

The W25Q128JVPIQ provides 128 Mbit (16 MiB) of external QSPI flash. It stores the boot image, application firmware, and any filesystem or data region defined by the selected board-package partition scheme. The BOOT control participates in the RP2350 USB boot workflow used to load recovery or update images.

The APS6404L-3SQR-ZR provides 8 MiB of external PSRAM with chip select on GPIO0. In the Arduino workflow, the board configuration must set PSRAM CS to GPIO0 and select the 8 MiB capacity. Static objects can be placed in PSRAM with the core's `PSRAM` attribute; dynamic objects can be allocated with `pmalloc()` and released with `free()`.

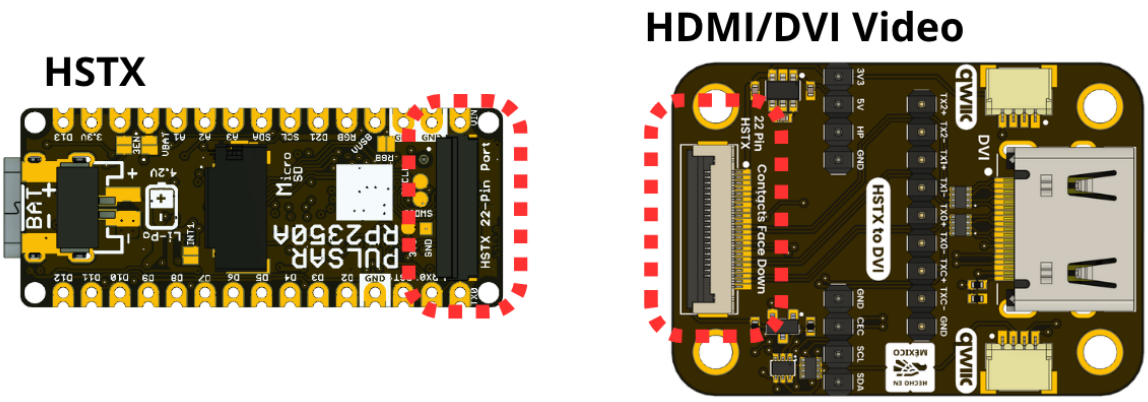



*External PSRAM location*

PSRAM is suited to framebuffers, file buffers, captured audio, sensor history, and large application structures. Interrupt flags, frequently accessed state, and timing-critical data should remain in internal SRAM. The application must check allocation results and avoid assuming that all configured PSRAM remains free after global/static allocations.

## 3.5 HSTX Video Output

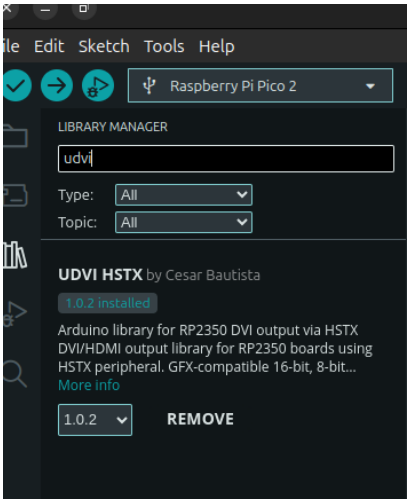
The HSTX peripheral serializes data at high speed and can generate the pseudo-differential TMDS signals used by DVI-compatible displays. The wiki and PCB routing define four channel pairs:




*HSTX connector location*

| TMDS channel | Positive GPIO | Negative GPIO | Function |
|---|---|---|---|
| Clock | 14 | 15 | Pixel/link clock pair |
| Data 0 | 18 | 19 | TMDS data channel 0 |
| Data 1 | 16 | 17 | TMDS data channel 1 |
| Data 2 | 12 | 13 | TMDS data channel 2 |

The 22-position HSTX connector carries these signals and returns. The wiki uses the UDVI HSTX library with a 320 × 240 RGB565 framebuffer, drawing primitives, text, and partial-screen updates. This resolution is an implementation profile from the wiki, not a limit of the connector or RP2350A.

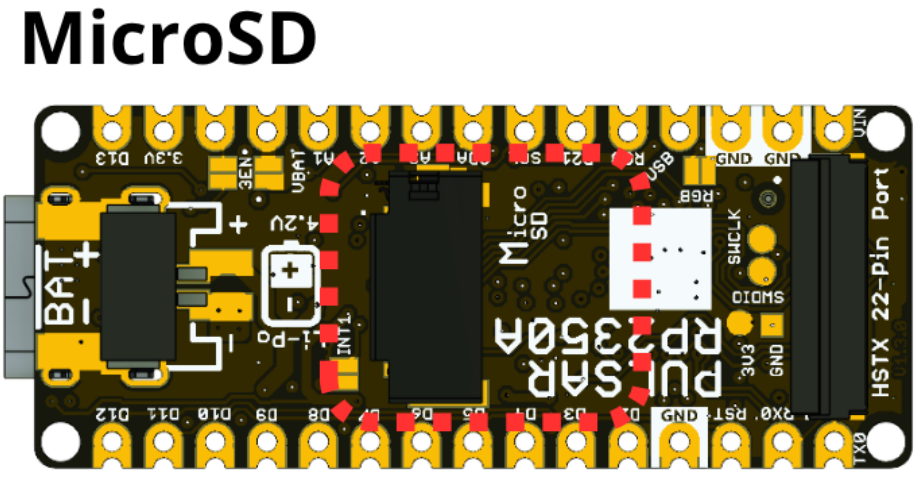



*UDVI HSTX library in Arduino Library Manager*

While HSTX is enabled, GPIO12–GPIO19 are actively driven and must not be shared with another output or an attached circuit that drives the same nets. Display adapters must match the connector contact order and differential-pair routing.

## 3.6 MicroSD Card Socket

The onboard 47309-2651 microSD socket is connected to a complete four-bit SDIO signal group and operates in the 3.3 V logic domain.




*microSD socket location*

| Signal | GPIO | SPI-mode role in current examples |
|---|---|---|
| SDIO_CLK | 2 | SCK |
| SDIO_CMD | 3 | MOSI |
| SDIO_DAT0 | 4 | MISO |

| Signal | GPIO | SPI-mode role in current examples |
|---|---|---|
| SDIO_DAT1 | 5 | Not used by SPI mode |
| SDIO_DAT2 | 6 | Not used by SPI mode |
| SDIO_DAT3 | 7 | Chip select |

The UNIT Electronics Wiki provides microSD examples using the SPI-compatible subset through SPI and SDFS, including card initialization, file creation and reading, directory enumeration, and data logging.

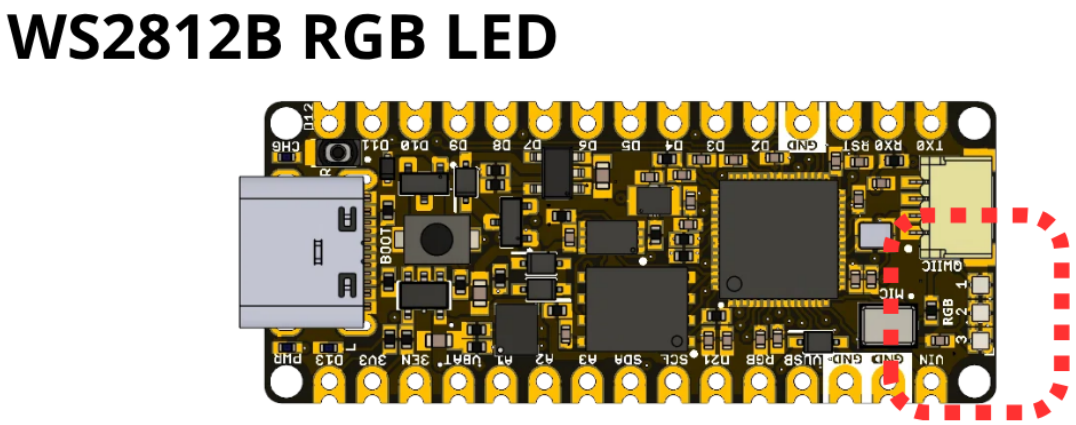
Software with four-bit SDIO support can additionally use SDIO_DAT1 and SDIO_DAT2, allowing the complete SDIO interface provided by the hardware to be used.

File writes should be flushed and closed before card extraction or power removal to reduce the risk of filesystem corruption.

Card capacity, speed class, and filesystem support depend on the software environment and microSD card used. FAT32 is recommended for the examples provided in the UNIT Electronics Wiki.

## 3.7 LED Indicators

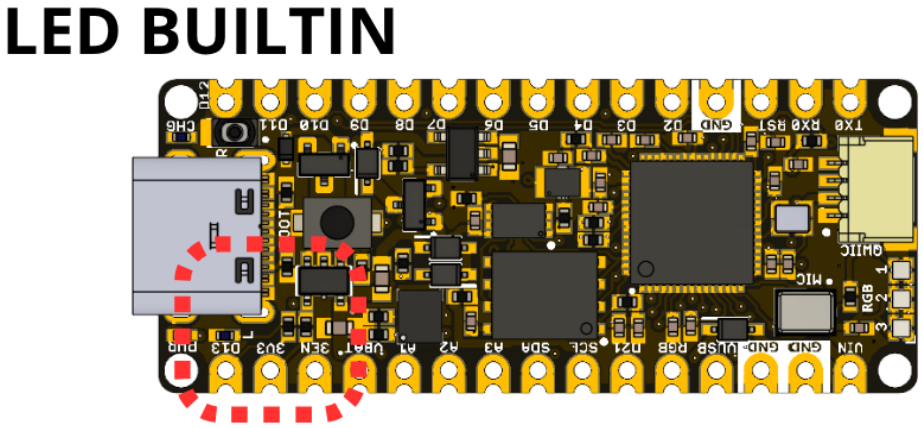
Three XL-1010RGBC-WS2812B addressable RGB LEDs form a serial chain driven by GPIO1. Firmware transmits an ordered color frame through the chain, allowing the three onboard pixels to provide status indication, progress animation, and user feedback from a single GPIO.




*WS2812 RGB LED locations*

The RGB_PAD solder jumper can connect the data output of the third onboard RGB LED to the castellated header. When this jumper is closed, the onboard RGB chain can be extended with compatible external addressable RGB LEDs.

The board also includes a user indicator on GPIO22 (D13 / BUILTIN1), a power indicator, and a battery charge-status indicator. The user LED and RGB LEDs are firmware-controlled. The power indicator follows the corresponding power rail, while the charge-status indicator is controlled by the MCP73831 STAT output.




*Built-in user LED location*

Applications should limit RGB brightness when power consumption or thermal rise is important. RGB LED control is independent of the HSTX video interface.

## 3.8 AP2112K and MCP73831 Power Management System

U1 is an AP2112K fixed 3.3 V LDO regulator. It converts the VSYS rail into the regulated 3.3 V logic domain used by the RP2350A, external memories, sensors, indicators, microSD interface, and QWIIC connector. The regulator supports up to 600 mA total output current, shared between the onboard circuitry and any external load connected to the 3.3 V rail.

The AP2112K EN signal controls the 3.3 V regulator. The corresponding castellated-header position is labeled 3EN on the PCB. The EN_PAD solder jumper connects the regulator EN signal to the 3EN castellated-header position, allowing external control of the 3.3 V regulator when the jumper is closed.

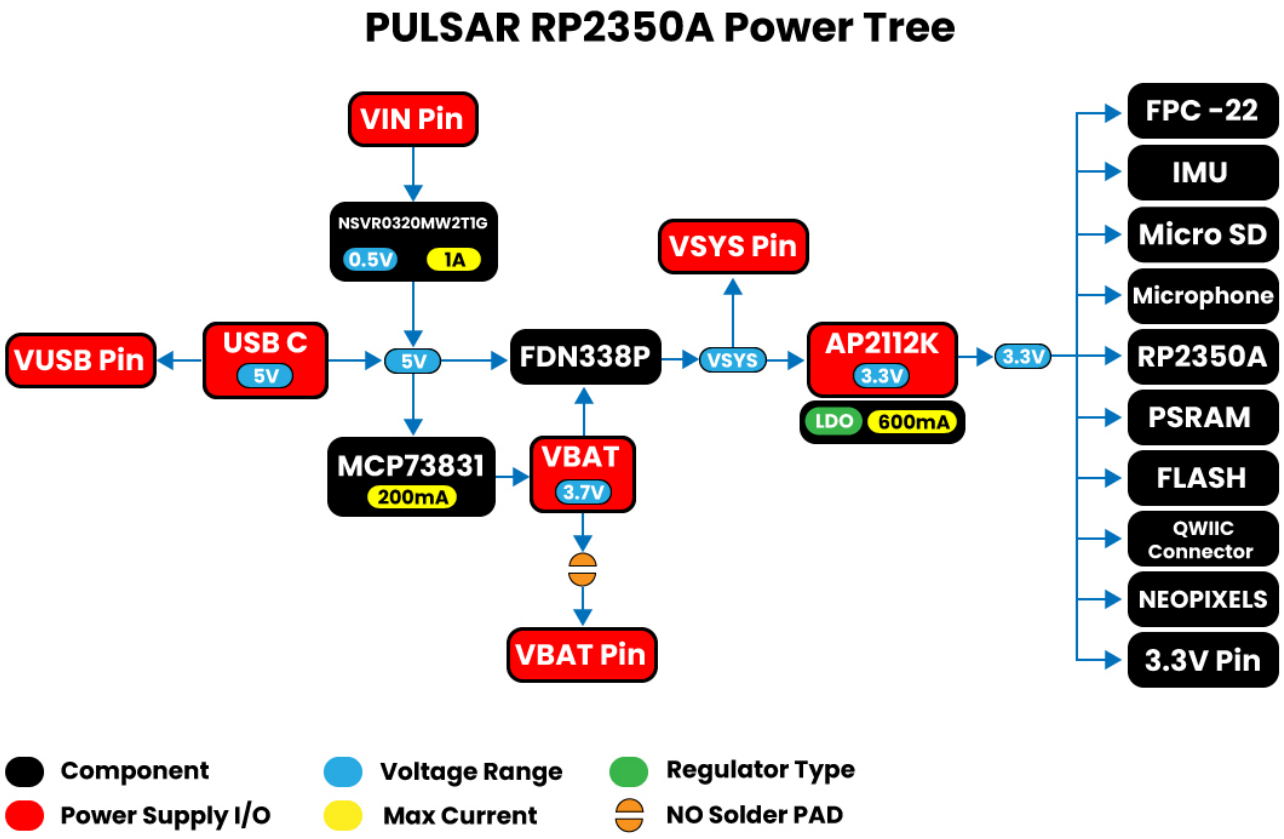
IC2 is an MCP73831T-2ACI/OT single-cell Li-Ion/LiPo battery charge controller. Hardware V1.3.0 is configured for a nominal charge current of 200 mA. The charge current is established by the PROG resistor and may be modified by changing this resistor according to the MCP73831 configuration requirements.

The battery input is intended for a single-cell LiPo battery with a nominal voltage of 3.7 V and a maximum fully charged voltage of 4.2 V. The battery path includes reverse-polarity protection.

The VBAT_PAD solder jumper allows the battery rail to be connected to the castellated header. When closed, the battery voltage can be accessed directly from the corresponding header position for external circuitry.

The board power-path circuitry combines USB, VIN, battery, VSYS, and the regulated 3.3 V domain. Schottky diodes and MOSFETs provide source routing and battery protection according to the implemented hardware design.

## 3.9 Power Tree




The power tree summarizes the main supply domains and their relationship within the UNIT PULSAR RP2350A. It is intended as a functional reference and does not replace the complete schematic.

The board can be powered from USB-C, VIN, or a single-cell LiPo battery. These sources are routed through the onboard power-path circuitry to the VSYS domain. The AP2112K-3.3 then generates the regulated 3.3 V rail used by the RP2350A and onboard peripherals.

The 3.3 V rail is also available through the castellated headers and QWIIC connector. The AP2112K provides up to 600 mA total output current. This current is shared between the RP2350A, memories, sensors, LEDs, microSD interface, and externally connected 3.3 V devices.

The current available to external loads therefore depends on the active onboard circuitry and application workload. The total 3.3 V load must remain within the regulator output capability.

The VBAT_PAD and EN_PAD solder jumpers provide optional access to the battery rail and regulator-enable control through the castellated headers.

## 3.10 BMI270 Motion Sensor

The onboard BMI270 combines a three-axis accelerometer and three-axis gyroscope. It is connected to GPIO8 (SDA) and GPIO9 (SCL) on the internal I2C bus. Firmware can detect the sensor through its supported I2C address configuration and read acceleration and angular-rate data.

The INT_PAD solder jumper provides an optional hardware interrupt connection between the BMI270 and GPIO12 of the RP2350A. When the jumper is closed, BMI270 interrupt events can be routed directly to GPIO12 for interrupt-driven applications without continuous polling.

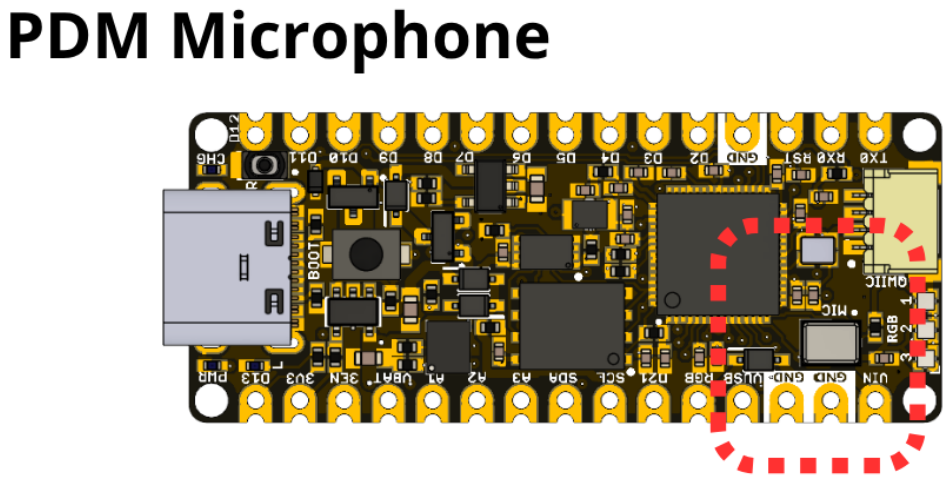
Typical applications include orientation interfaces, motion-triggered logging, gesture input, vibration monitoring, and motion-controlled graphics or user interfaces.

Measurement range, filtering, output data rate, interrupt source, and interrupt behavior are configured through the BMI270 registers or software driver and are not fixed by the PCB.

## 3.11 PDM Microphone

The ICS-41350 is an onboard digital MEMS microphone connected to the RP2350A through a PDM interface. The RP2350A supplies the PDM clock on GPIO10 and receives the microphone data stream on GPIO11.

The software PDM interface clocks the microphone and converts the PDM stream into PCM sample buffers for audio analysis, visualization, recording, streaming, or storage.




*PDM microphone location*

The microphone is mounted on the top side of the board, while its acoustic port is exposed through an opening in the PCB and is accessible from the bottom side.

The acoustic port must remain unobstructed. Enclosures, adhesive, dust, contamination, or mounting surfaces covering the port can affect acoustic performance.

Microphone sampling configuration and digital filtering are software-dependent. Final acoustic performance can also be affected by enclosure geometry and the mechanical integration of the board.

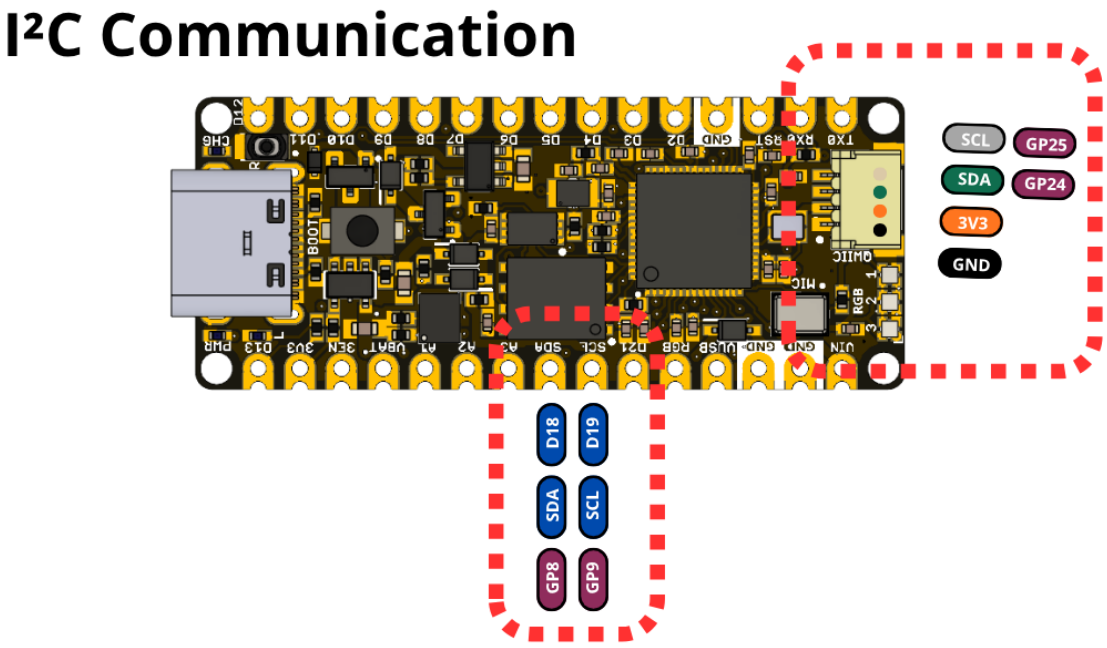
The UNIT Electronics Wiki provides software examples for PDM microphone initialization, audio acquisition, and processing.

## 3.12 I2C and QWIIC Expansion

The UNIT PULSAR RP2350A provides two independent I2C routes, allowing the onboard motion sensor and external QWIIC devices to operate on separate buses.

| Interface | SDA | SCL | Primary use |
|---|---|---|---|
| Internal I2C | GPIO8 | GPIO9 | Onboard BMI270 and castellated-header access |

| Interface | SDA | SCL | Primary use |
|---|---|---|---|
| QWIIC I2C | GPIO24 | GPIO25 | External QWIIC devices and HSTX connector access |




*I2C and QWIIC connections*

GPIO8 and GPIO9 connect the onboard BMI270 and are also exposed through the castellated headers. GPIO24 and GPIO25 connect to the four-position QWIIC connector and are also routed to the HSTX connector.

Both interfaces operate in the 3.3 V logic domain. External I2C devices must therefore be compatible with 3.3 V operation unless appropriate level translation is provided.

Using separate I2C routes allows applications to keep the onboard BMI270 bus independent from external QWIIC peripherals. Device addresses, pull-up configuration, bus capacitance, and cable length must be considered when multiple devices are connected to the same bus.

The UNIT Electronics Wiki provides additional examples for I2C bus scanning, BMI270 communication, EEPROM byte and block operations, hexadecimal dumps, and structured EEPROM data handling.

## 3.13 Combined System Operation

The UNIT PULSAR RP2350A is designed to support concurrent operation of its onboard subsystems. The RP2350A can combine sensing, audio acquisition, removable storage, external memory, RGB indication, I2C expansion, and HSTX video output within the same application.

For example, BMI270 motion data can be used to control HSTX-rendered graphics while PSRAM provides framebuffer or application storage. PDM microphone or I2C sensor data can be buffered in PSRAM and recorded to the microSD card, while the onboard RGB LEDs provide acquisition, storage, or system-status indication.

Applications using multiple high-bandwidth peripherals should consider memory allocation, DMA usage, interrupt timing, storage write latency, and peripheral resource sharing. Time-critical data and interrupt-related buffers should remain in internal SRAM where appropriate, while PSRAM can be used for larger framebuffers, audio buffers, sensor histories, and application data.

Shared GPIO functions must also be considered when enabling HSTX or optional solder-jumper connections.

Additional software examples in the UNIT Electronics Wiki demonstrate the individual subsystems before combining them into more complex applications.

# 4 Connectors & Pinouts

Pin labels describe hardware revision V1.3.0. GPIO numbers identify RP2350A signals; framework aliases depend on the selected board definition.

## 4.1 General Pinout

| Board label | RP2350A | Primary board role | Shared resource |
|---|---|---|---|
| TX0 / D1 | GPIO18 | UART-capable digital I/O | HSTX Data0 pair |
| RX0 / D0 | GPIO19 | UART-capable digital I/O | HSTX Data0 pair |
| D2 | GPIO17 | Digital I/O | HSTX Data1 pair |
| D3 | GPIO16 | Digital I/O | HSTX Data1 pair |
| D4 | GPIO15 | Digital I/O | HSTX clock pair |
| D5 | GPIO14 | Digital I/O | HSTX clock pair |
| D6 | GPIO13 | Digital I/O | HSTX Data2 pair |
| D7 | GPIO12 | Digital I/O | HSTX Data2 pair / optional BMI270 INT1 |
| D8 | USB_DM (pin 51) | USB D- | Shared with USB-C through 22 Ω series resistor R10 |
| D9 | USB_DP (pin 52) | USB D+ | Shared with USB-C through 22 Ω series resistor R11 |
| D10 / SS | GPIO21 | Digital I/O | General-purpose |
| D11 / MOSI | GPIO23 | Digital I/O | General-purpose |
| D12 / MISO | GPIO20 | Digital I/O | General-purpose |
| D13 / SCK / LED | GPIO22 | Digital I/O and user LED | BUILTIN1 |
| A1 / D15 | GPIO29 / ADC3 | Analog-capable I/O | Also routed to HSTX connector |
| A2 / D16 | GPIO27 / ADC1 | Analog-capable I/O | Castellated header |
| A3 / D17 | GPIO26 / ADC0 | Analog-capable I/O | Castellated header |
| SDA / D18 | GPIO8 | I2C data | BMI270 bus |
| SCL / D19 | GPIO9 | I2C clock | BMI270 bus |
| D21 / CLK_MIC | GPIO10 | PDM clock | ICS-41350 clock |
| DATA_MIC | GPIO11 | PDM data input | ICS-41350 data |
| RGB | GPIO1 | NeoPixel data input | Three onboard WS2812 LEDs |
| NEOPIXEL DOUT | — | RGB-chain extension output | Available at castellated header when RGB_PAD is closed |

Power and control positions include 3V3, 3EN, VBAT, 5V, VIN, RESET, and multiple GND connections.

3EN, VBAT, and NEOPIXEL DOUT are disconnected from their castellated-header positions by default to preserve the Nano-style pin layout. They can be enabled individually by closing the corresponding solder pads on the bottom side of the PCB.

The 3EN position provides access to the AP2112K EN signal when its corresponding solder jumper is closed, allowing external control of the 3.3 V regulator.

The INT_PAD solder jumper optionally connects BMI270 INT1 to D7 / GPIO12.

## 4.2 Arduino Nano Pinout Compatibility

The board uses two parallel 15-position castellated edge rows inspired by the Arduino Nano layout. This provides a familiar mechanical footprint and labeling pattern, but does not imply full electrical or shield compatibility.

Important differences include:

- The RP2350A and board GPIO operate in the 3.3 V logic domain.
- Several conventional positions have PULSAR-specific functions, including VBAT, 3EN, NEOPIXEL DOUT, and PDM microphone signals.
- 3EN, VBAT, and NEOPIXEL DOUT are disconnected from their castellated-header positions by default and can be enabled through bottom-side solder jumpers.
- D7 / GPIO12 can optionally receive the BMI270 INT1 signal by closing INT_PAD.
- GPIO12–GPIO19 are shared with the HSTX video interface.
- D8 and D9 carry the RP2350A native USB D-/D+ signals. They are connected to the same differential pair used by the USB-C connector through 22 Ω series resistors and should not be treated as conventional digital I/O.
- Analog channel labels do not follow a simple sequential GPIO order.
- Bottom-side connectors and components may require additional clearance when the board is installed in a carrier.

Before installing the board into a Nano-style carrier, verify power, reset, analog, digital, and mechanically shared positions, as well as bottom-side clearance.

## 4.3 QWIIC Connector

J1 is a four-position, 1 mm-pitch QWIIC connector carrying:

| Signal | RP2350A connection | Function |
|---|---|---|
| GND | Ground | Common return |
| 3.3 V | Regulated rail | Peripheral supply |
| SDA | GPIO24 | External I2C data |
| SCL | GPIO25 | External I2C clock |

The QWIIC interface operates in the 3.3 V logic domain and is separate from the GPIO8/GPIO9 I2C bus used by the onboard BMI270.

The QWIIC 3.3 V supply is provided by the same AP2112K regulator used by the board. The regulator supports up to 600 mA total output current, shared between the onboard circuitry and all external 3.3 V loads.

Use 3.3 V-compatible QWIIC peripherals and verify connector orientation before connection.

## 4.4 MicroSD Connector

| Socket signal | GPIO | Description |
|---|---|---|
| CLK | GPIO2 | SDIO clock / SPI SCK |
| CMD | GPIO3 | SDIO command / SPI MOSI |
| DAT0 | GPIO4 | SDIO data 0 / SPI MISO |
| DAT1 | GPIO5 | SDIO data 1 |
| DAT2 | GPIO6 | SDIO data 2 |
| DAT3 | GPIO7 | SDIO data 3 / SPI chip select |
| VDD | 3.3 V | Card supply |
| GND / shield | Ground | Electrical return and connector shield |

The microSD interface supports the complete four-bit SDIO signal group. Current UNIT Electronics Wiki examples use the SPI-compatible subset.

Insert or remove the microSD card only when filesystem activity has stopped. Software should flush and close open files before card extraction or power removal.

## 4.5 Battery Connections

JP1 provides the battery connection for a single-cell LiPo battery. Hardware V1.3.0 supports PH2.0 mm or PH1.25 mm battery connector options depending on the assembled configuration.

The battery input is designed for a single-cell LiPo battery with a nominal voltage of 3.7 V and a maximum fully charged voltage of 4.2 V. Battery charging is managed by the onboard MCP73831T-2ACI/OT and is configured for a nominal charge current of 200 mA.

Polarity markings are provided on the bottom side of the PCB. Always verify connector polarity before connecting a battery, as connector compatibility alone does not guarantee correct polarity.

The VBAT_PAD solder jumper can connect the battery rail to the corresponding castellated-header position. This connection is open by default. When the solder jumper is closed, the battery voltage becomes available at the VBAT castellated pad.

Do not connect multi-cell battery packs or batteries exceeding 4.2 V.

## 4.6 HSTX 22-pin Connector

J5 is a 22-position, 0.5 mm-pitch FFC/FPC connector that provides the RP2350A HSTX interface together with auxiliary I2C, analog, power, and ground connections.

| Pin | Signal | RP2350A connection | Function |
|---|---|---|---|
| 1 | D7 / INT1 | GPIO12 | HSTX Data2 / optional BMI270 INT1 |
| 2 | GND | Ground | HSTX return |
| 3 | D6 | GPIO13 | HSTX Data2 |
| 4 | D5 | GPIO14 | HSTX clock |

| Pin | Signal | RP2350A connection | Function |
|---|---|---|---|
| 5 | GND | Ground | HSTX return |
| 6 | D4 | GPIO15 | HSTX clock |
| 7 | D3 | GPIO16 | HSTX Data1 |
| 8 | GND | Ground | HSTX return |
| 9 | D2 | GPIO17 | HSTX Data1 |
| 10 | D1 | GPIO18 | HSTX Data0 |
| 11 | GND | Ground | HSTX return |
| 12 | D0 | GPIO19 | HSTX Data0 |
| 13 | A1 / D15 | GPIO29 / ADC3 | Analog-capable auxiliary signal |
| 14 | GND | Ground | Return |
| 15 | A0 | GPIO28 / ADC2 | Analog-capable auxiliary signal |
| 16 | GND | Ground | Return |
| 17 | SDA | GPIO24 | External I2C data |
| 18 | GND | Ground | I2C / auxiliary return |
| 19 | SCL | GPIO25 | External I2C clock |
| 20 | GND | Ground | I2C / auxiliary return |
| 21 | 3.3 V | Regulated 3.3 V rail | Peripheral supply |
| 22 | GND | Ground | Power return |

GPIO12 through GPIO19 provide the eight high-speed signals used by the RP2350A HSTX interface. In the DVI-compatible configuration, these signals form the four differential signal pairs required for video output.

The connector also exposes GPIO24 and GPIO25 for the external I2C bus, GPIO28 and GPIO29 as analog-capable auxiliary signals, and the regulated 3.3 V rail.

The UNIT DevLab DVI to FPC Adapter can be used with this connector to provide a DVI-compatible video interface.

When HSTX video output is active, GPIO12–GPIO19 must not be simultaneously driven by external circuitry through the castellated headers or other connections.

The INT_PAD solder jumper can connect BMI270 INT1 to GPIO12 (D7). Because GPIO12 is also part of the HSTX interface, applications using HSTX must account for this shared connection before closing INT_PAD.

Use a compatible 22-position, 0.5 mm-pitch FFC/FPC cable and verify cable contact orientation before connection. Remove board power before inserting or removing the FFC/FPC cable.

## 4.7 USB-C, BOOT, and Reset

J2 provides USB-C power, programming, and USB data connectivity. The USB D- and D+ signals connect to the RP2350A native USB PHY and are also exposed at the D8 and D9 positions through 22 Ω series resistors.

The BOOT button selects the RP2350A ROM USB boot mode used for firmware programming and recovery. The RESET button restarts the RP2350A while board power remains applied.

A typical firmware-recovery sequence is to hold BOOT while resetting or connecting USB, then release BOOT after the device enters USB boot mode.

## 4.8 SWD and Debug Pads

Bottom-side debug pads expose SWDIO, SWCLK, 3.3 V reference, and GND for connection to an external SWD debug probe. The SWD interface supports firmware programming and source-level debugging of the RP2350A.

**Silkscreen erratum:** The SWD VCC and GND labels are reversed. Refer to Section 9.6 for the corrected mapping before connecting a debug probe.

Use short debug connections and ensure that the target board and debug probe share a common ground. The 3.3 V pad should be used as the target-voltage reference for the debug probe and should not be used to power the complete board unless the probe and board power configuration explicitly support it.

# 5 Board Operation

The procedures below define the board workflow described by the technical wiki and the compilable sketches under `software/cpp_examples/`.

## 5.1 Getting Started with Arduino IDE

1. Install Arduino IDE 2.x.
2. Add the UNIT package index to **Additional Boards Manager URLs**: `https://raw.githubusercontent.com/UNIT-Electronics/Uelectronics-RP2040-Arduino-Package/main/package_Uelectronics_rp2040_index.json`
3. Install the UNIT Electronics RP2040/RP2350 package.
4. Select **UNIT Pulsar RP2350A** or an installed PULSAR-specific board definition.
5. Select the RP2350A chip variant, PSRAM CS GPIO0, and 8 MiB PSRAM.
6. Connect a USB-C data cable and select the enumerated serial/boot port.
7. Open the basic Blink sketch and upload it.

The repository examples were compile-checked with the UNIT RP2350 core using a 16 MiB flash and 8 MiB PSRAM configuration. Compile success confirms API and library compatibility; electrical operation also depends on the connected hardware and selected configuration.

## 5.2 First Power-up and Inspection

Use USB-C for initial power. Inspect component orientation, connector seating, solder bridges, and the microphone port before energizing the board. If available, use a current-limited source and monitor the 3.3 V rail.

Do not attach a battery, external VIN, HSTX adapter, microSD card, or QWIIC peripheral for the first power check. Confirm the power indicator and absence of unexpected heating, then establish USB enumeration and upload behavior.

## 5.3 BOOT and Reset Workflow

For a normal update, the selected framework may reset the board into its upload mode automatically. If automatic upload is unavailable:

1. Disconnect external driven signals.
2. Hold BOOT.
3. Press and release reset, or connect USB while BOOT is held.
4. Release BOOT after the ROM boot interface appears.
5. Copy or upload the generated firmware using the toolchain instructions.
6. Reset again to run the application from external flash.

SWD provides an alternative when debugging startup code or recovering a board whose USB application does not enumerate.

## 5.4 Recommended Bring-up Sequence

| Order | Subsystem | What the step establishes |
|---|---|---|
| 1 | User LED | Core execution, GPIO20, and upload path |
| 2 | USB serial | Host communication and diagnostic output |
| 3 | ADC | Basic analog input path |
| 4 | RGB chain | GPIO1 timing and three-pixel order |
| 5 | BMI270 I2C | GPIO8/GPIO9 and onboard sensor communication |
| 6 | QWIIC I2C | GPIO24/GPIO25 and external expansion |
| 7 | PSRAM | GPIO0 selection, capacity detection, and allocation |
| 8 | microSD | SPI-compatible subset, card access, and filesystem |
| 9 | PDM | GPIO10/GPIO11 clock and audio capture |
| 10 | HSTX | GPIO12–GPIO19 pairing and display output |
| 11 | Combined application | Memory ownership and subsystem coexistence |

Testing one subsystem at a time makes pin conflicts and library configuration errors easier to isolate.

## 5.5 C++ Example Collection

The `software/cpp_examples/` directory contains 28 complete sketches:

| Group | Sketches | Coverage |
|---|---|---|
| Blink | 2 | basic and serial-debug LED operation |
| ADC | 2 | continuous reading and threshold indication |
| WS2812 | 3 | color sequence, brightness, running pixel |
| I2C | 7 | scan, BMI270, EEPROM byte/block/dump/structure/multiple devices |
| microSD | 4 | initialize, read/write, list, log |
| HSTX | 3 | test pattern, drawing primitives, system monitor |
| Complete application | 1 | BMI270-controlled HSTX cube |
| PSRAM | 5 | detect, static/dynamic allocate, test, monitor |

| Group | Sketches | Coverage |
|---|---|---|
| PDM | 1 | capture PCM samples through the PDM library |

The examples reconcile known wiki/API differences: user LED GPIO20, SDFS capacity through `FSInfo`, directory open mode, `flush()` return type, and release of `pmalloc()` memory through `free()`.

## 5.6 Memory Planning

Keep stacks, interrupt state, frequently accessed variables, and small DMA descriptors in internal SRAM. Place large framebuffers, logging queues, and historical data in PSRAM. Check every allocation and track free internal and external heap separately.

HSTX RGB565 graphics can consume a large buffer; microSD and USB also require working memory. Avoid allocating every maximum-size buffer at startup without a documented memory budget.

## 5.7 Storage Operation

Configure the microSD SPI subset on GPIO2, GPIO3, GPIO4, and GPIO7 for the current SDFS examples. Initialize the card, verify filesystem information, open only the files needed, call `flush()` after important writes, and close handles before removal or reset.

For data logging, buffer short records in RAM and write them in bounded batches rather than performing filesystem operations from an interrupt handler.

## 5.8 Video, Sensor, and Audio Coexistence

HSTX reserves GPIO12–GPIO19 and may generate continuous DMA/interrupt load. BMI270 uses GPIO8/GPIO9, PDM uses GPIO10/GPIO11, and microSD uses GPIO2–GPIO7, so the fixed signal sets do not overlap. They still compete for processor time, DMA channels, SRAM, PSRAM bandwidth, and power.

Use bounded update rates, partial display redraws, buffered audio acquisition, and nonblocking storage tasks. A combined application should expose health information over USB serial or RGB status rather than failing silently.

## 5.9 Troubleshooting Guide

| Symptom | Checks |
|---|---|
| No USB device | Data-capable cable, BOOT/reset sequence, host port, 3.3 V rail |
| Blink does not run | GPIO20 mapping, selected board/FQBN, successful upload |
| PSRAM unavailable | PSRAM CS GPIO0, 8 MiB option, allocation result |
| BMI270 not found | GPIO8/GPIO9 configuration, address selection, library |
| microSD fails | FAT32, GPIO2/3/4/7, SDFS configuration, card seating |
| PDM returns no data | GPIO10 clock, GPIO11 data, PDM buffer/callback setup |
| HSTX display blank | pair order, adapter/cable orientation, supported timing |
| Unstable combined app | internal heap, PSRAM use, blocking loops, pin conflicts |

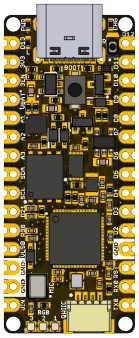
Record the exact board revision, package/core version, library versions, compiler output, power source, and reproduction steps for every bring-up issue.

# 6 Mechanical Information

Hardware revision V1.3.0 uses a narrow development-board form factor with two parallel castellated edge rows, four corner mounting holes, USB-C at one end, and the 22-pin HSTX FFC/FPC connector at the opposite end.

The board uses a 2.54 mm edge-pad pitch and a 15.24 mm spacing between the two castellated rows, providing a Nano-style mechanical layout.

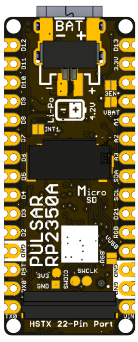
## 6.1 Component-side Envelope



The component side contains the USB-C connector, pushbuttons, QWIIC connector, RP2350A, external memories, BMI270 IMU, ICS-41350 PDM microphone, power-management circuitry, and other onboard components.

Adequate vertical and lateral clearance should be provided around the USB-C connector, pushbuttons, QWIIC connector, and other top-side components. The castellated edge pads must remain accessible when the board is mounted on headers, carrier boards, sockets, or directly soldered to another PCB.

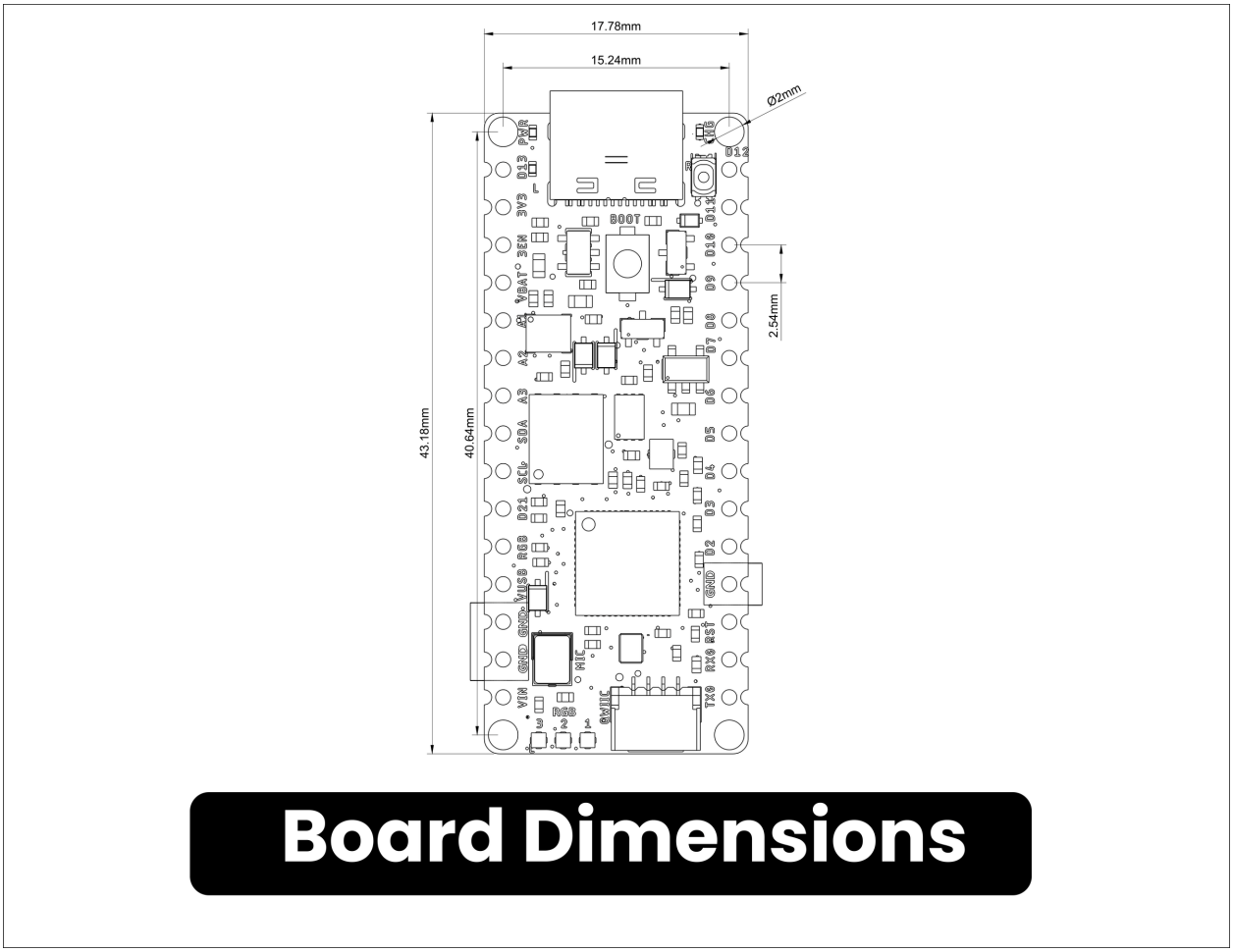
## 6.2 Bottom-side Envelope




The bottom side provides access to the battery connector, microSD socket, HSTX FFC/FPC connector, configuration solder pads, SWD debug pads, and the microphone acoustic port.

Adequate clearance must be maintained for battery connection, microSD card insertion and removal, and FFC/FPC cable installation. Enclosures and carrier boards must not obstruct the microphone acoustic port.

## 6.3 Board Dimensions




The principal mechanical dimensions of hardware revision V1.3.0 are summarized below.

| Parameter | Dimension |
|---|---|
| Overall board length | 43.18 mm |
| Overall board width | 17.78 mm |
| Castellated-row spacing | 15.24 mm |
| Castellated-pad pitch | 2.54 mm |
| Mounting-hole diameter | Ø2.0 mm |
| Vertical mounting-hole spacing | 40.64 mm |

Four mounting holes are located near the corners of the PCB. The castellated edge connections follow a 2.54 mm pitch, with 15.24 mm spacing between the two parallel rows.

The dimensional drawing should be used as the primary reference for board-outline and mounting dimensions when designing carrier boards, fixtures, or enclosures.

## 6.4 Mechanical Integration Considerations

When integrating the UNIT PULSAR RP2350A into a carrier board, fixture, or enclosure, provide sufficient clearance for:

- USB-C cable insertion and removal.
- QWIIC cable connection.
- Battery connector and cable routing.
- microSD card insertion and removal.
- HSTX FFC/FPC cable insertion and routing.
- BOOT and RESET button access.
- Castellated edge-pad soldering or header installation.
- SWD debug access when required.
- Configuration solder-pad access when required.
- The microphone acoustic port and its surrounding acoustic path.

PCB thickness, maximum top-side and bottom-side component heights, and connector insertion envelopes should be verified from the corresponding mechanical component specifications when they are critical to the final enclosure or carrier design.

# 7 Company Information

| Item | Information |
|---|---|
| Company | UNIT Electronics |
| Website | UNIT Electronics |
| Address | Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX, Mexico |

The repository and Product Wiki in the next chapter provide the product's hardware, software, and reference documentation.

# 8 Reference Documentation

The following resources provide the official hardware, software, and application documentation for the UNIT PULSAR RP2350A.

| Reference | Location |
|---|---|
| Product repository | UNIT-Electronics-MX/unit_pulsar_rp2350a |
| Hardware reference | hardware/README.md |
| Technical wiki | PULSAR RP2350A Wiki |
| Getting Started | Setup guide |
| C++ examples | software/cpp_examples |
| Board schematic | Hardware V1.3.0 schematic |
| RP2350 component datasheet | Raspberry Pi RP2350 Datasheet |

The UNIT Electronics Wiki provides application-oriented documentation, setup procedures, peripheral usage examples, and software workflows for the board.

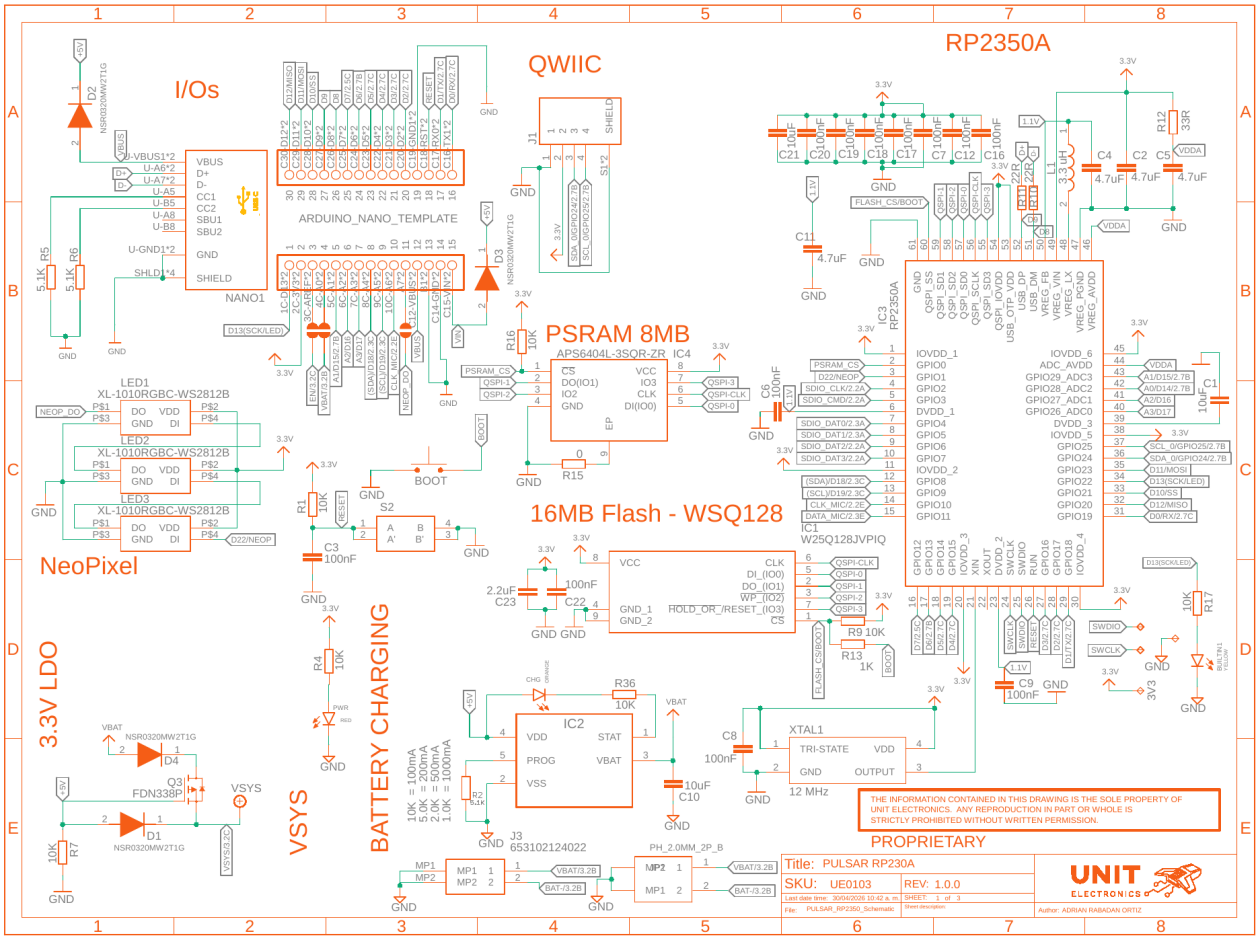
The hardware schematic and this Product Reference define the board-level implementation, connections, power architecture, interfaces, and hardware characteristics. RP2350A device-level electrical characteristics and processor specifications should be referenced from the official Raspberry Pi RP2350 datasheet.

# 9 Appendix

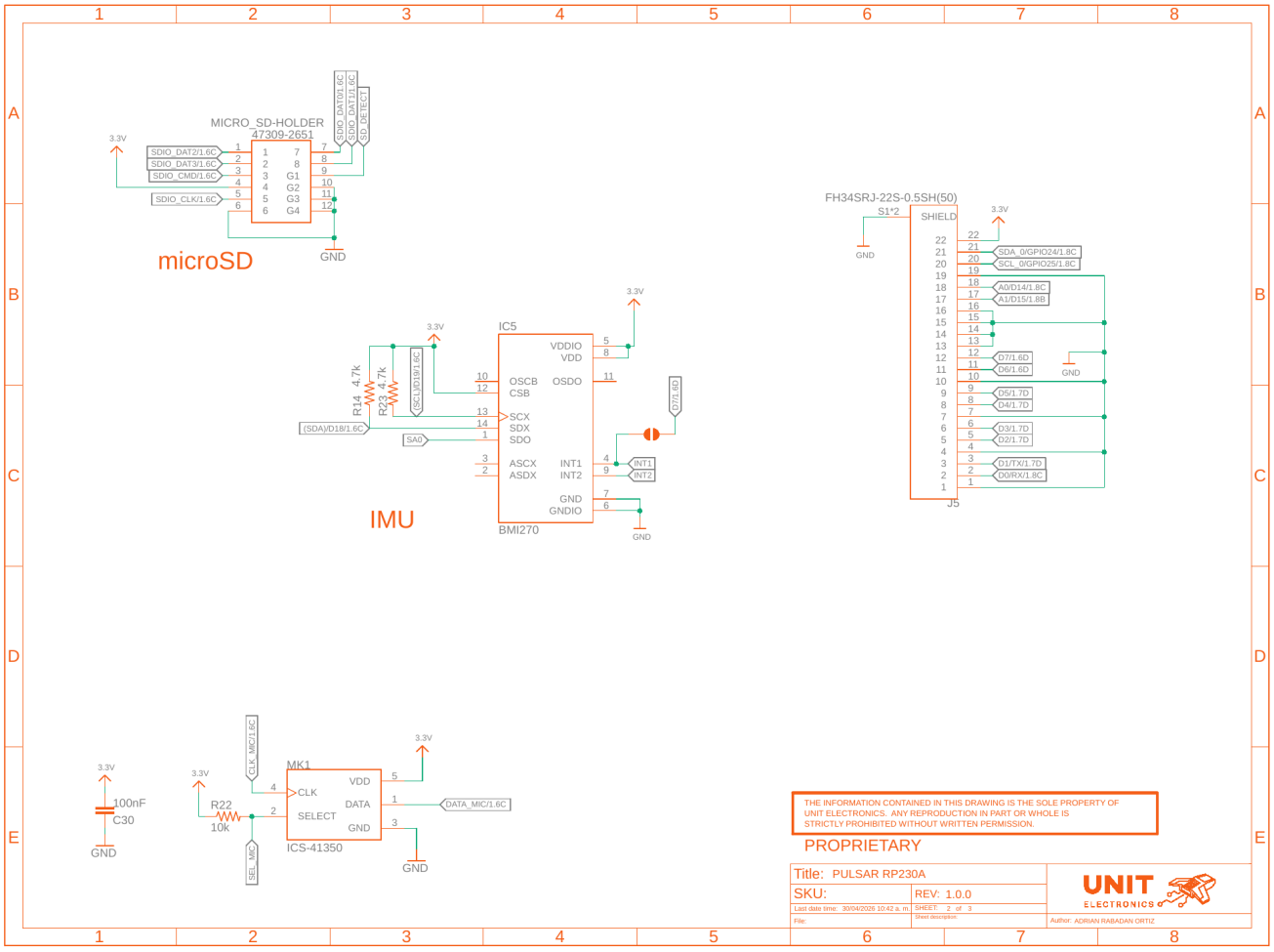
## 9.1 Schematic

The following three sheets are rendered from `unit_sch_v_1_3_0_pulsar_rp2350a.pdf`. The original PDF remains the authoritative electrical schematic for hardware revision V1.3.0.

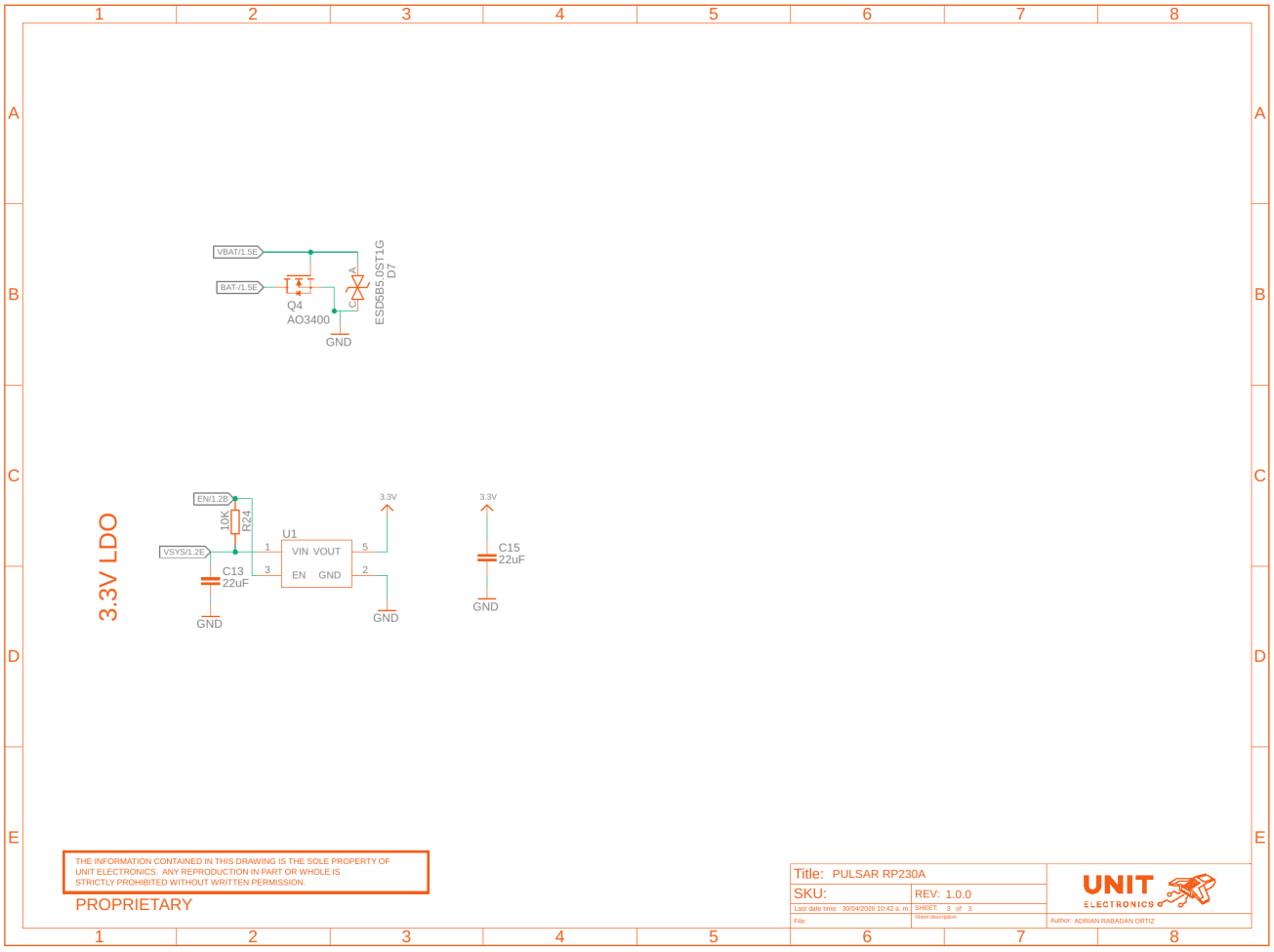
### 9.1.1 Processor, Memory, USB, I/O, and Power Entry



### 9.1.2 microSD, BMI270, HSTX Connector, and PDM Microphone



### 9.1.3 3.3 V Regulator and Battery Protection




## 9.2 Technical References

The documentation set for the UNIT PULSAR RP2350A is organized around the following technical sources:

1. The hardware V1.3.0 schematic defines board connectivity, component designators, power architecture, and electrical implementation.
2. The official Raspberry Pi RP2350 datasheet defines RP2350A device-level electrical characteristics, processor capabilities, GPIO behavior, ADC operation, and HSTX functionality.
3. The UNIT Electronics Wiki provides setup procedures, peripheral usage documentation, application examples, and software workflows.
4. The repository C++ examples provide implementation references for the board peripherals and supported software APIs.

## 9.3 Design and Application Notes

The following characteristics depend on the final application, firmware configuration, connected peripherals, or mechanical integration:

- Total board current consumption varies with processor activity, clock configuration, enabled peripherals, RGB LED brightness, microSD activity, PDM audio acquisition, PSRAM usage, HSTX video output, and external loads.
- Available current for external 3.3 V devices depends on the portion of the AP2112K 600 mA output capability consumed by the board itself.
- Maximum validated interface rates may depend on firmware configuration, connected devices, cable characteristics, and signal integrity.
- Final microphone acoustic performance depends on enclosure geometry, acoustic-port clearance, and mechanical integration.
- Thermal performance and maximum component temperatures depend on workload, ambient conditions, airflow, enclosure design, and external loading.
- Connector insertion clearance and maximum component-height requirements should be verified for the final carrier or enclosure design.

## 9.4 Document Control

| Field | Value |
|---|---|
| Product | UNIT PULSAR RP2350A |
| SKU | UE0103 |
| Product family | UNIT DevLab ecosystem |
| Hardware revision | V1.3.0 |
| Product Reference | Version 0.1.0 |
| Publication date | 2026-08-03 |

### 9.5 Source Notes

- The hardware V1.3.0 schematic title block contains the text PULSAR RP230A and revision 1.0.0. This Product Reference uses the released product identification UNIT PULSAR RP2350A and hardware revision V1.3.0.
- The microSD hardware exposes a complete four-bit SDIO interface. Current UNIT Electronics Wiki examples use the SPI-compatible CLK, CMD, DAT0, and DAT3 subset.
- D8 and D9 expose the RP2350A native USB D- and D+ signals through the board routing and must not be treated as conventional general-purpose digital I/O.
- D13 / BUILTIN1 is connected to GPIO22 on hardware V1.3.0.

## 9.6 Hardware Errata

### SWD VCC and GND Silkscreen Labels

On hardware revision V1.3.0, the VCC and GND silkscreen labels printed beside the SWD debug pads are reversed.

Use the following corrected electrical mapping when connecting an SWD debug probe:

| Printed silkscreen label | Actual electrical connection |
|---|---|
| VCC | GND |
| GND | 3.3 V reference |

Verify the SWD pad connections before connecting a debug probe.

Following the incorrect silkscreen labels may reverse the probe power-reference and ground connections and may damage the board or debug equipment.

Refer to Section 4.8 for SWD connection guidance.